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Split-Ensemble: Efficient OOD-aware Ensemble via Task and Model Splitting

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Abstract

Uncertainty estimation is crucial for deep learning models to detect out-of-distribution (OOD) inputs. However, the naive deep learning classifiers produce uncalibrated uncertainty for OOD data. Improving the uncertainty estimation typically requires external data for OOD-aware training or considerable costs to build an ensemble. In this work, we improve on uncertainty estimation without extra OOD data or additional inference costs using an alternative *Split-Ensemble* method. Specifically, we propose a novel *subtask-splitting* ensemble training objective where a task is split into several complementary subtasks based on feature similarity. Each subtask considers part of the data as in-distribution while all the rest as OOD data. Diverse submodels can therefore be trained on each subtask with OOD-aware objectives, learning generalizable uncertainty estimation. To avoid overheads, we enable low-level feature sharing among submodels, building a tree-like Split-Ensemble architecture via iterative splitting and pruning. Empirical study shows Split-Ensemble, without additional computational cost, improves accuracy over a single model by 0.8%, 1.8%, and 25.5% on CIFAR-10, CIFAR-100, and Tiny-ImageNet, respectively. OOD detection for the same backbone and in-distribution datasets surpasses a single model baseline by 2.2%, 8.1%, and 29.6% in mean AUROC, respectively.

1. Introduction

Deep learning models achieve high accuracy metrics when applied to in-distribution (ID) data. However, such models deployed in the real world can also face corrupted, perturbed, or out-of-distribution inputs (Hendrycks & Dietterich, 2019). Then, model predictions may not be reliable. Therefore, estimation of the epistemic uncertainty with OOD detection is crucial for trustworthy models (Gal & Ghahramani, 2016).

In general, uncertainty estimation is not a trivial task. Practitioners often consider various statistics derived from the uncalibrated outputs of softmax classifiers as confidence scores (Hendrycks et al., 2022). On the other hand, deep ensembling is another popular approach (Lakshminarayanan et al., 2017), where uncertainty can be derived from predictions of independently trained deep networks. However, deep ensembles come with large memory and computational costs, which grow linearly with the ensemble size. Recent research investigates strategies to share and reuse parameters and processing across ensemble submodels (Gal & Ghahramani, 2016; Wen et al., 2020; Turkoglu et al., 2022). Though these techniques reduce memory overheads, they suffer from the reduced submodel diversity and, thus, uncertainty calibration. Moreover, their computational costs remain similar to the naive deep ensemble since the inference through each individual submodel is still required.

More advanced methods for uncertainty estimation using a single model include temperature scaling, adversarial perturbation of inputs (Liang et al., 2017), and classifiers with the explicit OOD class trained by OOD-aware outlier exposure training (Hendrycks et al., 2018). However, as outlier exposure methods typically outperform other approaches that do not utilize external OOD-like data, OOD data distribution can be unknown or unavailable to implement effective outlier exposure training in practical applications.

This work aims to propose novel training objectives and architectures to build a classifier with a single-model cost, without external OOD proxy data, while achieving ensemble-level performance and uncertainty estimation. To avoid the redundancy of having multiple ensemble submodels learn the same task, we, instead, split an original multi-class classification task into multiple complementary subtasks. As illustrated in Figure 1, each subtask is defined by considering a subset of classes in the original training data as ID classes. Then, the rest of the training set is a proxy for OOD distribution for the subtask. Feature similarity is used

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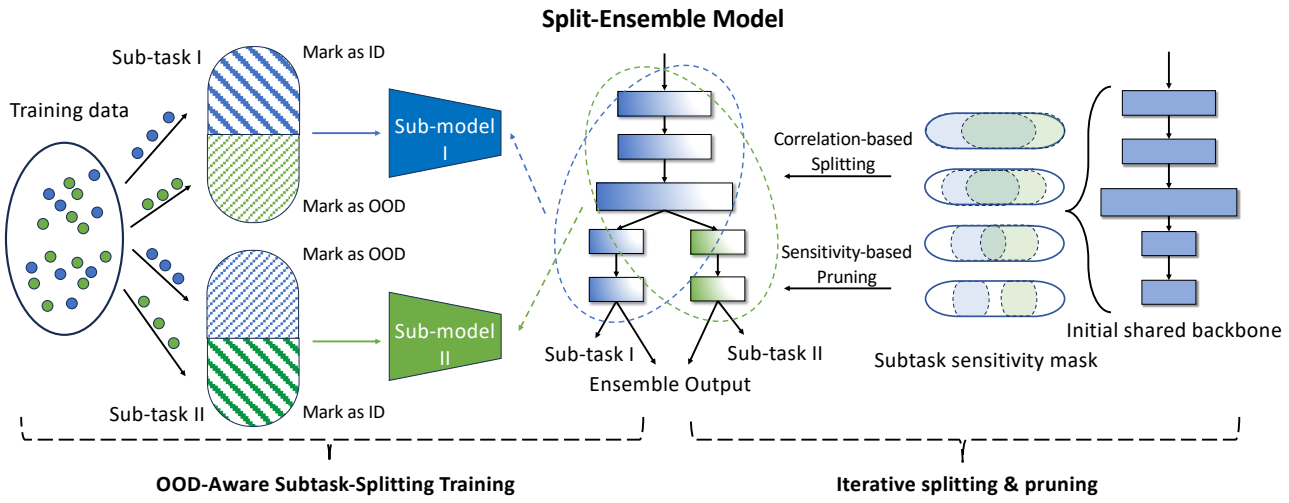

Figure 1. **Overview.** We split an original task into complementary subtasks to create objectives for submodel training. All submodels form an ensemble to perform the original task, and, importantly, each submodel can be trained with OOD-aware objectives of the subtask as proposed in Section 3. To implement an efficient Split-Ensemble architecture, we start with a shared backbone and iteratively perform splitting and pruning based on subtask similarity and sensitivity described in Section 4.

in grouping the subtasks so that the ID and OOD classes can be well separated. This enables our novel *training objective with subtask splitting*, where each submodel learns an OOD-aware objective for generalizable uncertainty estimation without external data. Finally, an ensemble of all submodels implements the original multiclass classification.

Our splitting objective requires a method to design computationally efficient submodel architectures for each subtask. Most subtask processing, as part of the original task, can utilize similar low-level features. Hence, it is possible to share early layers across submodels. Moreover, as each subtask is easier than the original task, we can use lighter architectures for the latter unshared processing in submodels when compared to the backbone design of the original task. Considering these two key observations, we propose a novel *iterative splitting and pruning* algorithm to learn a tree-like Split-Ensemble model. As illustrated in Figure 1, the Split-Ensemble shares early layers for all submodels. Then, they gradually branch out into different subgroups, resulting in completely independent branches for each submodel towards the last layers. Global structural pruning is further performed on all the branches to remove redundancies in submodels. Given the potential large design space of Split-Ensemble architectures, we propose correlation-based splitting and pruning criteria based on the sensitivity of model weights to each subtask's objective. This method enables automated architecture design through a single training run for our Split-Ensemble.

In summary, the paper makes the following contributions:

- We propose a subtask-splitting training objective for OOD-aware ensemble training without external data.
- We propose a dynamic splitting and pruning algorithm to build an efficient tree-like Split-Ensemble architecture corresponding to the subtask splitting.
- We empirically show that the proposed Split-Ensemble approach significantly improves accuracy and OOD detection over a single model baseline with a similar computational cost, and outperforms larger ensemble baselines by a factor of $4\times$.

In the following sections, we first discuss related work in Section 2, followed by deriving the subtask-splitting training objectives in Section 3, then propose criteria and algorithm for splitting and pruning mechanism in Section 4, and finally present experiment results in Section 5.

2. Related Work

2.1. OOD Detection and OOD-aware Training

OOD detection has a long history of research, including methods applicable to deep neural networks (DNNs). Lane et al. (2007) firstly propose to use classification confidence scores to perform in-domain verification through a linear discriminant model. Liang et al. (2017) enhance OOD detection by temperature scaling and adversarial perturbations. Papadopoulos et al. (2021) propose a method to explicitly make the model aware of OOD using an outlier-exposure training objective with OOD proxy data. Zhou et al. (2020)

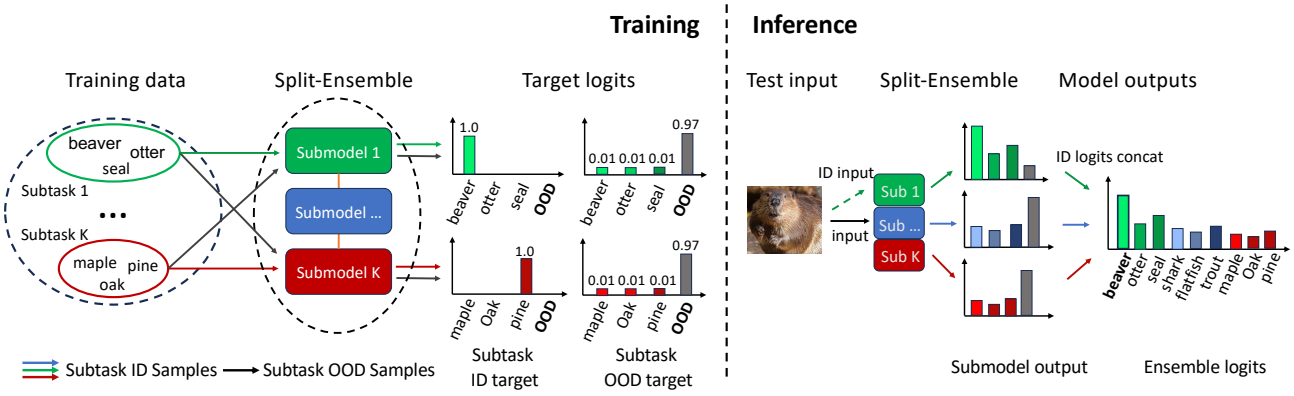

Figure 2. **Subtask splitting.** Each submodel learns its subtask using a subset of the original training data. OOD detection by outlier exposure training is realized using other subtasks' examples. Concatenated ID logits from all submodels implement the original multiclass classification task.

investigate the differences between OOD-unaware/-aware DNNs in model performance, robustness, and uncertainty. Jeong & Kim (2020) propose a few-shot learning method for detecting OOD samples. Besides supervised learning, Winkens et al. (2020); Sehwag et al. (2021) investigate self-supervised OOD detector based on contrastive learning. Wang et al. (2022) propose partial and asymmetric supervised contrastive Learning (PASCL) to distinguish between tail-class ID samples and OOD samples. However, the above single-model OOD detectors cannot be implemented without the availability of OOD proxy data. In contrast, our work proposes a novel subtask-splitting training objective to allow OOD-aware learning without external data.

2.2. Deep Ensembles

Ensemble methods improve performance and uncertainty estimation by using predictions of multiple members, known as submodels. Lakshminarayanan et al. (2017) propose the foundation for estimating uncertainty in neural networks using ensemble techniques. However, computational and memory costs grow linearly with the number of submodels in deep ensembles. To improve efficiency, Havasi et al. (2021) replace single-input and single-output layers with multiple-input and multiple-output layers, Gal & Ghahramani (2016) extract model uncertainty using random dropouts, and Durasov et al. (2021) utilize fixed binary masks to specify network parameters to be dropped. Wen et al. (2020) enhance efficiency by expanding layer weights using low-rank matrices, Turkoglu et al. (2022) adopt feature-wise linear modulation to instantiate submodels from a shared backbone, and Valdenegro-Toro (2023) ensembles only a selection of layers. These methods aim to mitigate the parameter overheads associated with deep ensembles. However, they cannot reduce computational costs because each submodel runs independently. Split-Ensemble overcomes the redundancy of ensemble processing by having submodels that run complementary subtasks with layer sharing. In addition, we further optimize Split-Ensemble design with tree-like architecture by splitting and pruning.

2.3. Efficient Multi-task Learning

With the subtask splitting, our method also falls into the domain of multi-task learning. Given the expense of training individual models for each task, research has been conducted to train a single model for multiple similar tasks. Sharma et al. (2017) propose an efficient multi-task learning framework by simultaneously training multiple tasks. Ding et al. (2021) design Multiple-level Sparse Sharing Model (MSSM), which can learn features selectively with knowledge shared across tasks. Sun et al. (2021) introduce Task Switching Networks (TSNs), a task-conditioned architecture with a single unified encoder/decoder for efficient multi-task learning. Zhang et al. (2022) develop AutoMTL that automates efficient MTL model development for vision tasks. Sun et al. (2022) propose a pruning algorithm on a shared backbone for multiple tasks. In this work, we explore correlations between multiple subtasks to design a novel splitting and pruning algorithm. Previous works on split-based structure search consider layer splitting (Wang et al., 2019; Wu et al., 2019; 2020) to increase the number of filters in certain layers for better model capacity. Split-Ensemble, on the other hand, uses architecture splitting as a way of deriving efficient architecture under a multi-task learning scenario of subtask-splitting training, leading to a novel tree-like architecture.

3. Subtask Splitting Method

In this section, we provide the definition of subtask splitting given a full classification task in Section 3.1. We derive